

TrES-1b

R-0242

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_250623 · created 2026-07-17T05:24:02+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460849.8515 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.172 +/- 0.011
Transit depth (Rp/Rs)^2	2.97 +/- 0.38 [%]
Orbital Inclination (inc)	89.93 +/- 0.54
Airmass coefficient 1 (a1)	1.056 +/- 0.01
Airmass coefficient 2 (a2)	-0.052 +/- 0.056
Scatter in the residuals of the lightcurve fit is	0.97 %
Best Comparison Star	#1 - [516, 432]
Optimal Method	PSF photometry
Transit Duration (day)	0.1067 +/- 0.0013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.172 ± 0.011 vs 0.1358 (deviation 3.29σ)
Duration fit vs published	0.1067 d vs 0.1042 d (deviation 1.95σ)
Mid-time O–C	-3.1 min
Depth SNR	15.6
Residual scatter	0.97 %

Light curve

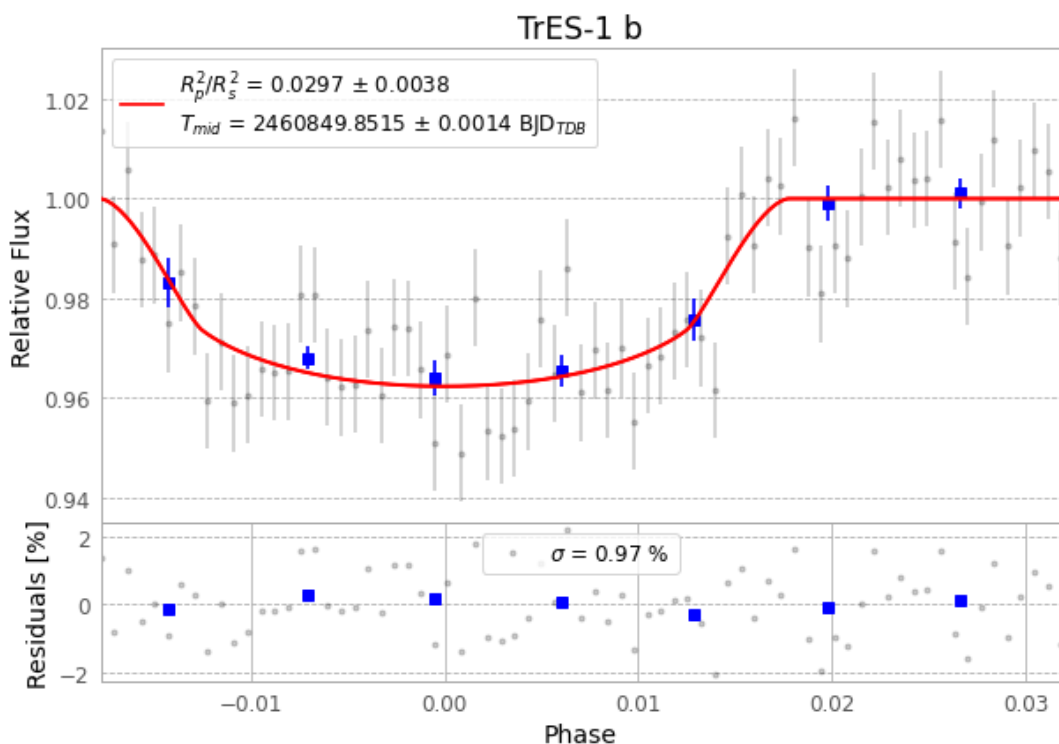


Figure 1 — FinalLightCurve_TrES-1 b_2025-06-23.png (SHA-256 7b24c0f4424a05f8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [341, 251], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 6371611fe6456df2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 825eb7e

Data provenance

73 FITS frames (48 MB), TRES-1250623070315.FITS → TRES-1250623103917.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-250623.log (e3c4acdb2a60...), FinalLightCurve_TrES-1 b_2025-06-23.png (7b24c0f4424a...), FinalLightCurve_TrES-1 b_2025-06-23.csv (4f50c08e6b67...)

Compute resources

Wall time 150.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 05:24Z.